

GEO5017 Final Project: Waste Detection in the Urban Environment

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1. Problem Statement

In the past decades, smart cities has been one of the most relevant concepts in the field of urban planning and management. This concept refers to the use of increasing amounts of data about the urban environment and of new technologies that allow efficient processing of this data to improve the offer and quality of urban services. In this context, this project explores how this concept can be applied to a specific problem: the monitoring of street cleanliness, an important aspect for urban up-keeping and for the quality of life of a city’s inhabitants.

The primary objective of this project is to implement an object detection workflow capable of identifying irregularly discarded waste within street view imagery. Specifically, the task involves developing a pipeline to process a dataset of $N = 10,000$ unannotated images from the city of Amsterdam. By integrating a modern deep learning algorithm (YOLO) with spatial data, the project aims to refine a model that generalizes effectively across varying urban contexts. The ultimate goal is to maximize the Precision at 100 (P@100) for the project leaderboard. This is achieved by isolating the Top 100 detections in the test set where the model exhibits its highest predictive confidence, and providing an automated solution for urban cleanliness assessment.

As an additional goal, we have implemented a second model that is trained to detect the waste objects within each image and classify these instances into four distinct categories: bulky waste, cardboard, garbage bags, and litter. These categories are described in Table 1. This second model outputs the bounding boxes of each waste object, alongside their respective label. The Top 100 predictions were also isolated according to the model’s confidence score, and the results are analysed separately in Section 4.2

2. Methodology Justification

2.1 Data Split

Following the given data split strategy, images from 2016-2019 were assigned to the training set, while those from 2020-2021 formed the validation set. The final evaluation was performed on the test dataset, which combined data from 2022-2023, and had a total of 4,092 images. This guaranteed a 6:4 train-test ratio.

2.2 Data Annotation

The provided 10,000 street view images were manually annotated based on two criteria. Firstly, the images were labelled as **waste** or **no_waste**. In second place, the waste objects were individually identified and labelled according to four predefined urban waste categories (Table 1). These two separate annotation methods were used to create the most appropriate inputs for each of the models; the Classify model benefited from the simpler labelling format, that also improved its computation efficiency, while the Detection model required the definition of bounding boxes for its implementation and evaluation.

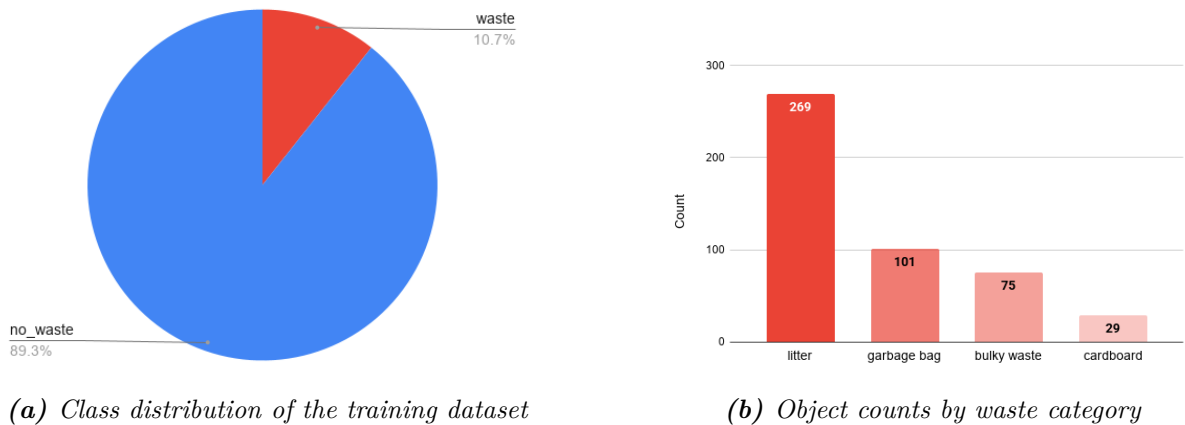
Table 1: Definition of urban waste categories

Class	Description
Bulky waste	Large discarded objects that cannot not fit into standard municipal waste bins (e.g., furniture, large appliances) and large concentrations of waste (e.g. piles with more than three garbage bags)
Garbage bag	Waste contained with standard, tied plastic garbage bags
Cardboard	Discarded cardboard materials, including intact or flattened packaging boxes
litter	Small pieces of materials improperly scattered on the ground (e.g., plastic bag, tissues, cans)

Two characteristics were observed from the annotated training dataset:

- **Class imbalance:** As shown in Figure 1, there is a clear imbalance not only between the `waste` or `no_waste` classes, but also among the individual waste categories.
- **Prevalence of small waste objects:** Litter accounts for 56.7% of all waste instances with an average area of 488.43 squared pixels (Table 2), meeting the MS COCO definition for small objects ($area < 32 \times 32$ pixels) (Lin et al., 2014).

Figure 1: Statistics of the training dataset.



2.3 Model Selection

The development of automated urban monitoring systems has been significantly accelerated by the advent of real-time object detection frameworks. Principal among these is the YOLO (You Only Look Once) architecture, which introduced a paradigm shift by treating object detection as a single regression problem, from image pixels to bounding box coordinates and class probabilities.

Table 2: Average bounding box pixel areas for each waste class in the training dataset.

Class Name	Average BBox Area (px ²)
litter	488.43
garbage bag	5,454.95
bulky waste	24,119.31
cardboard	5,861.96
Total Average	5,614.52

According to Xiuling et al., 2024, YOLOv3-Final model outperforms other state-of-the-art detectors (e.g., Fast R-CNN and Faster R-CNN) in small object detection. Consequently, given the characteristic of our training dataset, we selected the YOLO model for this task, specifically adopting its latest version (YOLO26n) to maximize detection capability.

At the same time, we were aware that some limitations of the model, described by Redmon et al., 2016 and Sapkota and Karkee, 2025, could have a negative impact on its performance for this project. According to the aforementioned authors, the YOLO model struggles with detection of overlapping objects, which could compromise the detection of piles of waste. Additionally, it may struggle to generalize to objects in new or unusual aspect ratios or configurations (Redmon et al., 2016). As our training set contains mostly small litter objects, the model could present poorer performance for under-represented waste categories.

3. Workflow

3.1 Labelling

Firstly, the 10,000 images from the designated dataset were manually labelled with the aid of open data annotation platform CVAT. The annotation consisted of two steps. The images were labelled with tags **waste** or **no_waste**, and the three data splits (training, validation, and test) were individually exported in the Ultralytics YOLO Classification 1.0 format, to be used by the Classify model. In sequence, we defined the bounding boxes of waste objects and labelled them according with the previously mentioned waste categories. The data splits were exported as Ultralytics YOLO Detection 1.0 files; this format is proper for the implementation of the Detection model. The definition of the "ground true" bounding boxes was also relevant for the computation of the confidence score of this model, as will be described in Section 4.2.

3.2 Classify model

To identify images containing waste, we fine-tuned a pretrained YOLO26's Classify model (YOLO26n-cls), which classifies an entire image into one of predefined classes. The algorithm consisted of three steps: training, validation, and prediction.

Handling class imbalance Initially, we trained the model using the entire training dataset. However, due to the class imbalance in the dataset as described in Section 2, the recall rate for the **waste** class was 0%, meaning the model simply classified every test image as **no_waste**. To address this, we did two experiments: applying class weights and

undersampling. First, taking advantage of customizable Ultralytics training pipeline, we modified the class weights in the loss function based on the number of instances per class, namely, upweighted the underrepresented class. This aimed to penalize misclassifications on the rare class more heavily, implemented as follows:

```
class_instance_counts = np.array([2534, 309])
class_weights = max(class_instance_counts) / class_counts
```

However, the model’s performance remained insufficient, achieving only a 1% recall rate for the **waste** class. Next, we applied random undersampling to the **no_waste** class. By implementing a code script, 2,236 **no_waste** images were randomly selected and removed to achieve a 1:1 class ratio. Training on this undersampled dataset ($N = 608$) yielded improved results, increasing the recall rate for the **waste** class to 11%. Therefore, this undersampled dataset was used for training the final model.

Hyperparameter tuning During the training process, empirical testing indicated that at least 100 epochs were required to produce meaningful results. Thus, we set the maximum number of epochs to 300 with an early stopping patience of 20 epochs to halt training when performance plateaued, thereby preventing overfitting. We also attempted further hyperparameter tuning (e.g., adjusting **scale**, **translate**, and **flipud**) using the YOLO **tune()** module. However, due to hardware constraints, we were unable to run a sufficient number of epochs and iterations to obtain optimal tuning results. As a result, we decided to retain the default hyperparameters, modifying only the epoch and patience settings.

The final model was evaluated on the test dataset. The model processes the unseen image to predict their respective classes and ranks them based on the prediction confidence for the **waste** class. To assess the model’s performance, we generated a confusion matrix, calculated the Precision at 100 (P@100) score, and extracted the top 100 images classified as **waste** along with their confidence annotations. The detailed results and evaluations are discussed in Section 4.1.

3.3 Object Detection model

Following the manual annotations performed in CVAT, a second model was implemented to localize and categorize waste objects within images by outputting bounding boxes alongside specific category labels. The model utilized the four predefined urban waste categories to provide an assessment on street cleanliness. The training pipeline followed a temporal split where images from 2016-2019 dataset were assigned to the training set while the 2020-2021 dataset served as the validation set for hyperparameter tuning.

Handling class imbalance & hyperparameter tuning As with the Classify model, while we initially attempted to automate tuning via the YOLO **tune()** module, the high computational demand exceeded our hardware capabilities, leading to system crashes. Consequently, we performed manual tuning to handle class imbalance and small object detection. Specifically, we set **cls=2.0** to increase the weight of correct class predictions and **mosaic=0.5** to assist the model in recognizing smaller instances within images.

The final model was evaluated on the test dataset using a custom automated leaderboard function. This script processed the test imagery to extract detections and ranked unique images based on the highest predictive confidence of their primary waste object. To verify the results, the pipeline performed an automated comparison between the model’s top predictions and the manually defined “ground truth” names. This allowed us to calculate the final P@100 score and export 100 annotated visual samples, where

each bounding box and label were plotted alongside its confidence score to facilitate the qualitative assessment described in Section 4.2.

4. Performance Analysis

4.1 Classify model

As shown in Table 3, the Classify model (YOLO26n-cls) performs well in identifying clean streets (**no_waste**) with a high accuracy rate of 97%, whereas the metrics regarding waste image identification require improvement. The false negatives and recall rates for the **waste** class are 89% and 11%, respectively. This indicates that despite undersampling, the model failed to fully generalize the distinct features of diverse types of wastes across varied environments. Meanwhile, the $P@100$ score, which represents the proportion of actual waste within the model’s top 100 most confident waste predictions, is 0.60. This score implies partial success in feature learning. While the model can reliably detect waste instances that have clear visual features (Figures 2a, 2b, and 2c), it also misclassifies non-waste objects that visually resemble waste (Figure 2d) and clean streets (Figures 2e and 2f) as waste. This misclassification is presumed to be due to the loss of background diversity caused by undersampling.

Table 3: Confusion Matrix of the ‘yolo26n-cls’

		True Label	
		no_waste	waste
Predicted Label	no_waste	0.97 (2819)	0.89 (146)
	waste	0.03 (81)	0.11 (18)

Figure 2: Top-Ranked True Positive and False Positive Samples Based on Confidence



Figure 3: Qualitative results from the top 100 ‘waste’ predictions. The top row shows True Positives where the model correctly identified waste and the bottom row shows False Positives where the model misclassified clean streets as ‘waste’.

4.2 Object Detection model

The Object Detection model (YOLO26n) achieved an overall mAP@0.5 of 0.176. As shown in Table 4, litter was the best performing class with mAP of 0.345 and the highest precision at 0.440. In contrast, cardboard proved the most difficult to detect, with mAP of 0.013 (Figure 5b).

Table 4: Object Detection Performance Metrics (YOLO26n)

Class	Images	Instances	Box(P)	R	mAP50
All classes	4092	976	0.232	0.234	0.176
Litter	311	510	0.440	0.355	0.345
Garbage bag	177	282	0.296	0.230	0.162
Bulky waste	97	103	0.194	0.350	0.182
Cardboard	60	81	0.000	0.000	0.013

In the Normalized Confusion Matrix (Figure: 4), we can identify background interference as the primary source of error. The model failed to detect 76% of litter and 84% of

garbage bags, misclassifying them as background. This is likely due to class imbalance between waste and no-waste images during the training phase of the model, or due to items blending into complex urban textures like pavement. Despite this, the accuracy between categories was high. Once the object was localized, the model rarely confused waste types (e.g. only 1% cross-class error between garbage bags and litter).

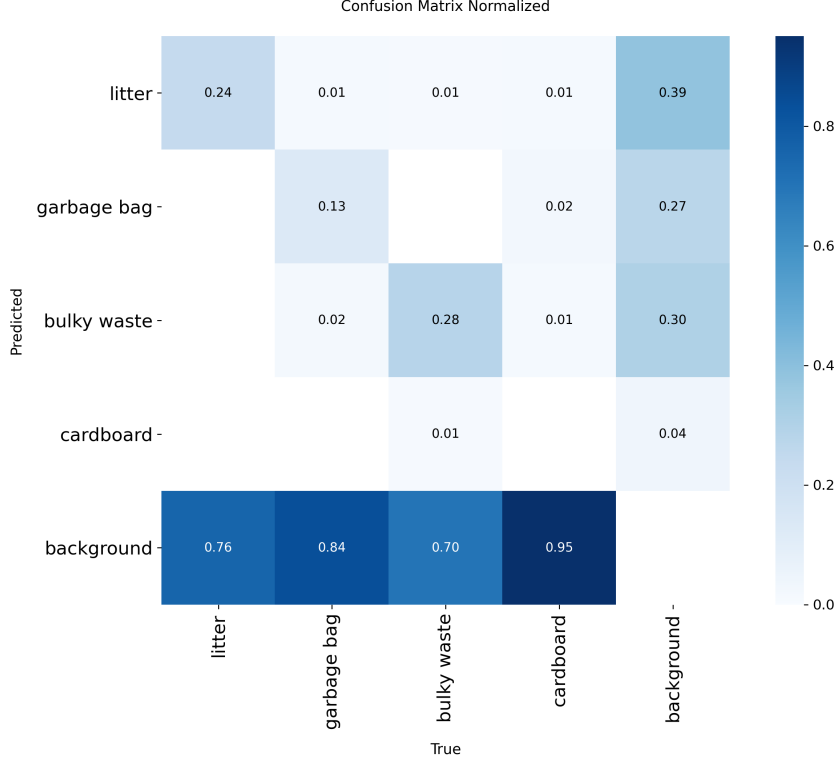
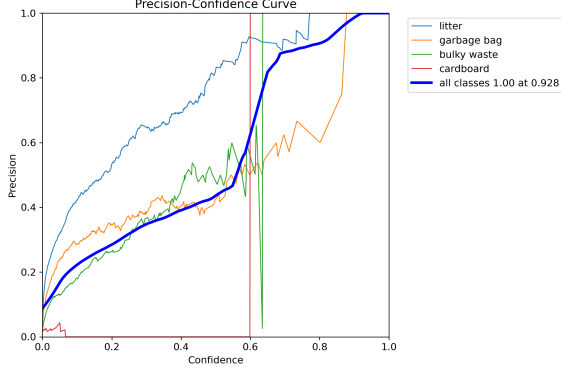


Figure 4: Normalized Confusion Matrix for the YOLO26n Detection Model. The matrix illustrates significant background interference, with 76% of litter and 84% of garbage bags being misclassified as background.

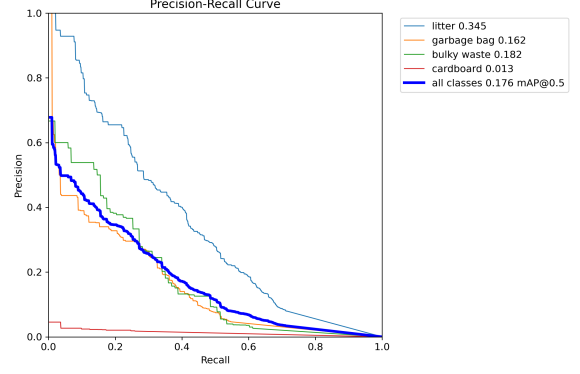
Our strategy to maximize P@100 is supported by the Precision-Confidence Curve (Figure 5a), which shows precision rising sharply as the confidence threshold increases. The model reaches a perfect precision of 1.00 at a 0.928 confidence threshold. Specifically, out of the Top 100 predictions generated by the automated leaderboard script, 83 instances were correctly identified as true waste objects based on the ground-truth comparison (test dataset). By isolating the Top 100 highest confidence detections, our pipeline effectively bypassed background-related false positives to provide highly certain results for the leaderboard. The F1-Confidence Curve (Figure 5c) confirms this trade-off, showing that while overall balance peaks at a low 0.109 confidence, the “high-precision” tail we targeted ensures maximum predictive reliability.

For this model, the confidence is computed based on two aspects: the confidence that the bounding box contains an object, and the accuracy of the predicted box (Redmon et al., 2016). This second element is defined as the intersection over union (IOU) of the predicted and true bounding box (Equation 1). Thus, the high confidence scores achieved by the model’s Top 100 detections represent not only its ability to detect the objects, but of also accurately identifying the boundaries of the objects’ bounding boxes.

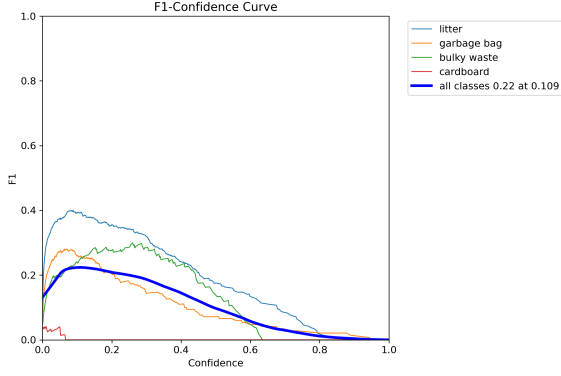
$$IOU = \frac{predicted \cap true}{predicted \cup true} \quad (1)$$



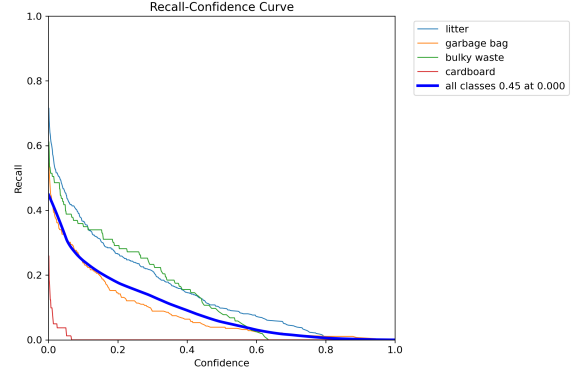
(a) Precision-Confidence Curve



(b) Precision-Recall Curve



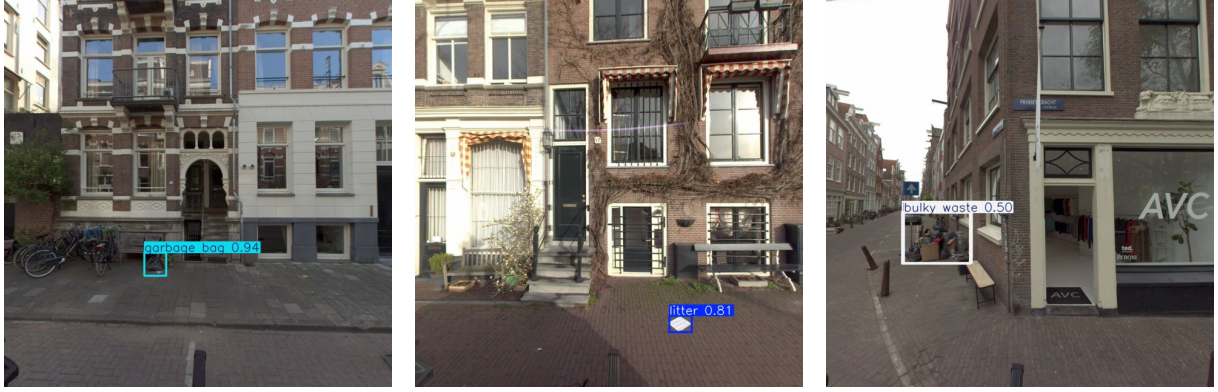
(c) F1-Confidence Curve



(d) Recall-Confidence Curve

Figure 5: YOLO26n Performance Curves. The Precision-Confidence curve (a) demonstrates that a precision of 1.00 is reached at a 0.928 threshold, justifying our P@100 strategy. The PR curve (b) shows the class-specific mAP values, highlighting the gap between litter (0.345) and cardboard (0.013).

Figure 6: Qualitative Analysis of the Top 100 Object Detection Predictions



(a) Garbage Bag (Conf: 0.94)

(b) Litter (Conf: 0.81)

(c) Bulky Waste (Conf: 0.50)



(d) Background (Conf: 0.58)

(e) Unlabelled litter (Conf: 0.56)

(f) Background (Conf: 0.50)

Figure 7: Qualitative results from the detection model Top 100. The top row illustrates True Positives where the model successfully localized waste instances with bounding boxes. The bottom row highlights False Positives where urban background textures were misidentified as waste.

5. Experiment

Code

- **GitHub** (<https://github.com/juliafmarques23/geo5017-group-05/tree/main/A3>)

6. Work Distribution

- Chaeyeon Moon (6477453):
 - Data labeling: years 2016 and 2017;
 - Code: Image classification using YOLO26n-cla model;
 - Report: Section 2, 3.2, 4.1;
- Evangelia Palli (6435939):
 - Data labeling: years 2018 and 2019;
 - Code: Object detection using YOLO26n model;
 - Report: Section 2, 3.3, 4.2;
- Julia Fossa Marques (6550975):

- Data labeling: years 2020 and 2021;
- Code: Object detection using YOLO26n model;
- Report: Section 1, 2, 3.1, 4.2;

References

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- Sapkota, R., & Karkee, M. (2025). Ultralytics yolo evolution: An overview of yolo26, yolo11, yolov8, and yolov5 object detectors for computer vision and pattern recognition. *idk yet*. <https://arxiv.org/html/2510.09653v2#S5>
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